

Integrated Signal-Processing and Self-Supervised Learning Frameworks for Low-Light Enhancement in Advanced Driver-Assistance Systems

The operational integrity of Advanced Driver-Assistance Systems (ADAS) is fundamentally predicated on the quality of environmental perception derived from on-board visual sensors. In ideal conditions, contemporary deep learning architectures for object detection and semantic segmentation exhibit high reliability; however, these systems face a catastrophic performance degradation when subjected to low-light environments typical of nighttime driving.¹ The core of this challenge lies in the physical limitations of RGB camera sensors, which, when operating in photon-starved conditions, suffer from a significantly reduced signal-to-noise ratio (SNR), limited dynamic range, and non-uniform illumination artifacts.³ Traditional methods for low-light image enhancement (LLIE) often fail to balance the competing requirements of noise suppression and detail preservation, while supervised deep learning models are constrained by the scarcity of high-quality paired training data in dynamic driving scenarios.⁵ Consequently, the research frontier has shifted toward self-supervised frameworks that leverage signal-processing priors to guide the enhancement process without the need for external ground-truth supervision.⁷

Theoretical Foundations of Visual Degradation and Sensor Physics

Understanding the degradation of visual signals at night requires an examination of the interaction between light sources and camera sensor electronics. When incident light intensity decreases, the number of photons reaching the sensor diminished, leading to a decline in SNR as defined by photon transfer theory.⁴ This decline manifests as amplified dark current noise and readout noise, which are exacerbated when cameras attempt to compensate for low illumination by increasing ISO sensitivity or extending exposure times.⁴ For ADAS, these physical constraints are not merely aesthetic issues but safety-critical obstacles; the loss of high-frequency structural information—such as object contours, traffic light signals, and ground markings—directly increases the probability of collision.³

The visual system in autonomous vehicles must interpret scenes where illumination is not only low but also non-uniform. Nighttime urban environments are characterized by high-intensity localized light sources, such as vehicle headlights and streetlamps, juxtaposed with extremely dark shadows.⁵ This creates a high dynamic range (HDR) problem where standard linear sensors saturate in bright regions while losing all signal in dark regions. Modern

signal-processing assisted enhancement attempts to model these complexities using physical priors, most notably the Retinex theory, which posits that an image is the product of a reflectance component, representing the intrinsic properties of objects, and an illumination component, representing the light falling upon them.¹²

The Evolution of Low-Light Enhancement Architectures

The progression of LLIE research has moved through several distinct phases, starting from classical pixel-wise transformations to sophisticated self-supervised neural architectures.

Classical Signal Processing and Histogram Techniques

Early attempts at low-light enhancement relied on statistical adjustments of the image histogram. Global Histogram Equalization (GHE) and Contrast Limited Adaptive Histogram Equalization (CLAHE) were developed to enhance contrast by redistributing pixel intensities.¹² While computationally efficient, these methods are notorious for amplifying noise and causing local information loss, as they lack the capacity to distinguish between signal and background noise.¹² Furthermore, they often produce unnatural color shifts and halo artifacts around high-contrast edges, which can confuse downstream ADAS perception algorithms.

Supervised Deep Learning and Paired Data Bottlenecks

The advent of Convolutional Neural Networks (CNNs) allowed for end-to-end mapping between low-light and normal-light images. Models such as Retinex-Net and MBLLN demonstrated that deep networks could learn to decompose images and adjust illumination more effectively than hand-crafted algorithms.¹⁴ However, supervised learning requires paired datasets—low-light images and their perfectly aligned normal-light counterparts—which are exceptionally difficult to capture in the real world.⁵ Datasets like LOL (Low-Light) provide static pairs, but they do not reflect the dynamic motion and complex lighting of a nighttime highway.¹⁷

Architecture Category	Representative Models	Training Data Requirement	Primary Constraint
Traditional	CLAHE, HE, Retinex (SSR/MSR)	None (Heuristic-based)	Noise amplification; poor color constancy ¹²
Supervised Deep	Retinex-Net, LLNet, MBLLN, LLE_UNET	Paired images (Low/Normal)	Lack of real-world generalization; high cost ⁵
Unsupervised GAN	EnlightenGAN, CycleGAN	Unpaired images	Training instability; potential for hallucinations ⁸
Self-Supervised	Zero-DCE,	Single low-light images	Semantic-perceptual

	LightenFormer, UMFNet	only	disconnect; over-exposure ⁷
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The Rise of Self-Supervised and Zero-Reference Learning

To overcome the data bottleneck, zero-reference learning models like Zero-DCE reformulate the enhancement problem as an image-specific curve estimation task.⁷ Instead of pixel-to-pixel mapping, the network learns to predict a series of high-order curves that adjust the dynamic range of the input image.⁷ This process is guided by a set of non-reference loss functions—including spatial consistency, exposure control, color constancy, and illumination smoothness—which measure the quality of the enhanced image without needing a ground-truth reference.⁷ While Zero-DCE is highly efficient and suitable for real-time applications, it struggle with extremely non-uniform lighting and can lead to over-enhancement of already bright areas, such as headlights.²³

Signal-Processing Assistance: Bridging Physics and Machine Learning

The integration of traditional signal-processing techniques into deep learning frameworks—known as signal-processing assistance—provides the inductive bias necessary to handle the complex degradations of low-light scenes. This is achieved through frequency-domain analysis, specialized decomposition, and physical priors.

Wavelet-Domain Decomposition and Noise Decoupling

One of the most effective signal-processing strategies in contemporary LLIE is the use of the Discrete Wavelet Transform (DWT). Unlike standard spatial convolutions, wavelets can decompose an image into multi-frequency components, typically one low-frequency (LL) sub-band and three high-frequency (LH, HL, HH) sub-bands.⁴ In a low-light context, the LL sub-band captures the global illumination and structural contours, while the high-frequency bands contain the textures and most of the sensor noise.¹

Networks like WTEFNet and LightenResDiff leverage this decomposition to implement a "divide-and-conquer" strategy.⁴ The low-frequency components are processed by an enhancement module focused on brightness recovery, while the high-frequency components are handled by a denoising or refinement module to suppress artifacts without blurring edges.⁴ This approach is particularly advantageous for ADAS because it preserves the sharp gradients required for edge detection while removing the "salt-and-pepper" noise that often triggers false positives in object detectors.¹

Retinex-Based Iterative Decomposition

Modern frameworks have evolved beyond simple Retinex decomposition to iterative models like RSD-Net and RetinexDIP.⁸ These models treat the decomposition of reflectance and illumination as a generative or iterative optimization problem. For instance, RetinexDIP uses

the internal statistics of a single image to estimate the illumination map, weakening the coupling between light and reflection to prevent noise from leaking into the final output.⁸ In ADAS applications, this decoupling is essential for maintaining color constancy under varying streetlamp spectral distributions.³

S-Curve and Gamma Transformation Priors

Signal processing also provides non-linear mathematical priors for light adjustment. Gamma correction, a classic signal-processing operation, is often integrated into deep networks (as seen in VELIE) to provide a decoupled initial enhancement.³ More advanced models, such as LightenFormer, utilize S-curve estimation networks (SCENet) to adaptively adjust the dynamic range.²¹ The "invertibility of the S-curve" is even used as a self-supervised loss function, ensuring that the enhancement process is mathematically stable and reversible.²¹

Signal Processing Component	Function in LLIE Framework	Impact on ADAS Performance
Discrete Wavelet Transform (DWT)	Separates structure from noise ⁴	Improved edge detection; reduced false positives ¹
Retinex Decomposition	Decouples reflectance and illumination ¹²	Better color constancy; glare modeling ¹³
Gamma/S-Curve Priors	Provides non-linear brightness mapping ³	Stable illumination across diverse scenes ³
Maximum Entropy Prior	Constrains reflectance entropy ²⁰	Prevents detail loss; maintains contrast ²⁰

ADAS-Specific Requirements: Real-Time Inference and Edge Deployment

Unlike general-purpose image enhancement, ADAS enhancement must satisfy stringent real-time constraints and operate on resource-constrained embedded hardware. The latency of the enhancement module directly affects the total reaction time of the autonomous system.

Hardware Performance and Latency Benchmarks

Research has demonstrated that even lightweight models must be optimized for deployment on edge GPUs like the NVIDIA Jetson series. For example, the LLE_UNET architecture achieves state-of-the-art results but its deployment on a Jetson Nano requires quantization (FP16/INT8) and conversion to optimized formats like TensorRT or TensorFlow Lite.¹⁷

Model	Parameter Count	Latency (Jetson Nano/Orin)	PSNR/SSIM (LOL)
LLE_UNET	Lighter than CycleGAN ¹⁷	~20 FPS (optimized) ¹⁷	28.71 / 0.82 ¹⁷

VELIE	Specialized Lightweight ³	0.19 s (per frame) ³	Competitive ³
Lightweight Framework	~1.3 M parameters ¹⁹	~6 FPS (Jetson Nano) ¹⁹	28.4 / 0.88 ¹⁹
WTEFNet	Embedded Platform Optimized	Real-time on Orin ⁵	State-of-the-art ⁵
UMFNet	Frequency-Guided ¹	High Efficiency ¹	28.14 / 0.88 ¹

The computational cost of self-supervised methods is often lower during inference because they do not require the complex multi-stage architectures of supervised models like MIR-Net, which, despite high quality, are often too slow for real-time driving. The trend toward "zero-shot" learning allows for a network that is both fast and robust to new environments without retraining.²⁶

Downstream Task Integration: The Semantic Link

A critical insight from recent research is that visual quality metrics like PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index) do not always correlate with object detection accuracy.⁹ An image might appear "bright" and "clean" to a human, but if the enhancement process has smoothed over the features used by a YOLO or Faster R-CNN detector, the safety of the vehicle is compromised.⁹ Consequently, modern self-supervised frameworks are increasingly incorporating "task-level" supervision.⁹

Experimental results show that integrating an advanced model like WTEFNet or UMFNet can improve mAP (mean Average Precision) on datasets like ExDark or DarkFace by significant margins—up to 15-20% compared to traditional methods.¹ This is achieved by designing loss functions that specifically penalize the loss of semantic information, often by using a frozen pre-trained detection model as a secondary supervisor.⁹

High Dynamic Range Challenges: Headlight Glare and Suppression

A primary failure mode for existing self-supervised models in driving scenarios is the presence of localized overexposure. Most zero-reference models assume the entire image needs brightening, which results in "blooming" artifacts around streetlights and headlights, obscuring pedestrians who might be standing near these light sources.¹⁷

Dual-Illumination and Glare Modeling

Zero-DiDCE addresses this by introducing a dual-illumination input, processing both the original image and its inverted version.²³ By learning from the inverted image, the model can identify areas that are already bright (which become dark in the inversion) and apply a "suppression" operation rather than an "enhancement" operation.²³ Similarly, Retinex-MEF extends the standard Retinex model to explicitly account for glare effects, using a shared reflectance component across multi-exposure frames to ensure that glare does not distort

the estimated object properties.¹³

Gated Positional Attention for Glare Suppression

New architectures like NRGS-Net (Nighttime Road Glare Suppression Network) employ gated positional attention (GPA) modules.¹¹ These modules integrate positional encoding with local contextual information to accurately locate and suppress glare regions.¹¹ This localized approach is superior to global gamma correction because it can dim high-beam headlights of oncoming vehicles without darkening the rest of the road view, maintaining visibility of hazards in the shadows.¹¹

Temporal Stability and Video Consistency in ADAS

For a vehicle moving at high speed, consistency between consecutive frames is vital. Single-frame enhancement methods often suffer from "temporal flickering," where brightness levels jump sporadically between frames due to varying sensor noise levels, leading to unstable object detection and "ghosting" artifacts.⁶

Spatial-Temporal Consistent Decomposition

Recent breakthroughs in Low-Light Video Enhancement (LLVE) focus on spatial-temporal consistent decomposition.³⁰ These frameworks assume that while the illumination (I) changes frame-by-frame, the reflectance (R) of the physical objects should remain constant.³⁰ By calculating cross-frame correspondences and uncertainty values, the model links predictions across frames.⁶ A spatial smoothness loss is applied to the illumination maps, reflecting the physical reality that natural lighting fields are continuous over time.³⁰

Recurrent Refinement and Optical Flow

To ensure temporal continuity, models like Zero-TIG and DWTA-Net use recurrent refinement modules guided by optical flow.⁶ The temporal feedback module warps the enhanced result of the previous frame to align with the current frame, recursively incorporating past knowledge into the current enhancement pipeline.⁶ While optical flow is computationally expensive, downsampling strategies are often used to accelerate the process while still mitigating noise and flickering.⁶

Deep Insight: Identifying the Research Novelty Gap

Based on a comprehensive review of the literature from 2021 to 2025, there is a clear trajectory toward more "aware" self-supervised systems. However, several critical gaps remain that are highly suitable for a bachelor's thesis.

Gap 1: The "Spectral-Semantic" Trilemma in Real-Time Video

Current wavelet-based models (WTEFNet, UMFNet) are primarily single-frame and do not account for the temporal propagation of frequency-domain artifacts.¹ There is a gap in

developing a **Temporal Wavelet Loss** that ensures high-frequency sub-bands (where noise is concentrated) are consistent across frames without blurring moving objects.¹ A researcher could work on a "Learnable Wavelet Temporal Filter" that adapts its threshold based on the ego-vehicle's velocity and lighting conditions.

Gap 2: Foundation Model Collaborative (FoCo) Self-Supervision for Edge Devices

While the FoCo framework leverages multiple foundation models (SAM, CLIP, DINO) to guide enhancement, these models are too large to run in the loop during the enhancement process for real-time ADAS.⁹ A significant novelty gap exists in **Distilling Semantic Priors** from large foundation models into the self-supervised loss functions of a lightweight student network (like Zero-DCE or LLE_UNET).⁹ This would allow the student network to benefit from the "global understanding" of a foundation model without the inference-time cost.

Gap 3: Physics-Informed Glare-Shadow Equilibrium

Most models either focus on brightening (Zero-DCE) or suppressing glare (NRGS-Net).⁷ There is a lack of a unified, self-supervised framework that handles the **Shadow-Glare Equilibrium** using a physics-informed model of light scattering (Atmospheric Scattering Model) integrated with Retinex theory.¹¹ A thesis could explore using the "Dark Channel Prior" (DCP) not just for dehazing, but as a local constraint to identify shadows that need brightening vs. bright regions that need glare-reduction.³⁶

Proposed Research Strategy for the Thesis

The most promising "novelty gap" for a bachelor thesis on "Signal-Processing Assisted Self-Supervised Low-Light Enhancement for ADAS" lies at the intersection of **Wavelet decomposition, High-Dynamic Range Glare Suppression, and Semantic Stability**.

The DS-WGC (Dual-Scale Wavelet Glare-Consistency) Framework

A proposed novel architecture would involve a three-pronged approach:

1. **Dual-Scale Wavelet Decomposition:** Utilizing DWT to separate the input into a low-frequency illumination map and high-frequency structural maps.⁴
2. **Inversion-Guided Glare Suppression:** Applying the dual-illumination principle (from Zero-DiDCE) specifically within the low-frequency wavelet sub-band to selectively suppress headlights while brightening the road.²³
3. **Semantic Flow Loss:** Introducing a loss function that ensures the high-frequency wavelet coefficients of identified "critical objects" (pedestrians, vehicles) are preserved across frames, using a lightweight optical flow estimate or feature-matching similarity.⁶

This research would address the "spectral trilemma" ¹ while remaining within the computational budget of an embedded ADAS system.³ By leveraging the mathematical robustness of wavelets and the zero-reference flexibility of deep curve estimation, the student can produce a model that is inherently more robust to the "real-world edge cases"

that current state-of-the-art models struggle with.

Contextualizing Results within the Global ADAS Market

The drive for higher levels of autonomy (SAE Level 3 and Level 4) necessitates a shift from human-viewable enhancement to "perception-viewable" enhancement.² Major automotive players are increasingly moving away from expensive active sensors like high-definition LiDAR toward "vision-only" or "vision-first" systems.² In this context, the ability to extract reliable environmental data from a standard RGB sensor in the middle of a rainy, moonless night is not just a feature—it is the prerequisite for safety.

The proposed "Signal-Processing Assisted" approach offers a way to maintain the interpretability and reliability of traditional engineering systems while harnessing the transformative power of deep learning. As research moves into 2026 and beyond, the integration of these hybrid models will likely become the standard for all intelligent vehicle platforms, ensuring that the car's "eyes" never fail, regardless of the lighting conditions.

Synthesis of Core Insights for Professional Implementation

The following table summarizes the strategic insights derived from the synthesized research material, providing a roadmap for the proposed thesis work.

Research Domain	Key Finding	Strategic Implementation Insight
Architectural Bias	U-shaped encoder-decoder (UNet) with attention is the standard for spatial detail. ¹⁷	Integrate Convolutional Block Attention Modules (CBAM) between encoding/decoding paths. ¹⁷
Noise Management	Standard spatial denoising blurs semantic edges. ¹	Use Wavelet (DWT) decomposition to isolate and remove noise in high-frequency bands. ⁴
Exposure Control	Global adjustments fail in HDR scenes with headlights. ¹⁰	Use dual-illumination (inverted inputs) and gated attention for localized suppression. ¹¹
Temporal Stability	Frame-by-frame processing causes flickering. ⁶	Implement spatial-temporal consistent decomposition based on reflectance-constancy. ³⁰
Optimization	PSNR/SSIM do not equate to detection accuracy. ⁹	Use Task-Level losses derived from frozen perception models

Final Conceptual Framework for the Thesis

The proposed research topic, "Signal-Processing Assisted Self-Supervised Low-Light Enhancement for ADAS," should conclude its review by articulating a vision where the vehicle's enhancement network is not a standalone pre-processor but a "context-aware" front-end for the perception engine. The novelty lies in the fusion: using **Signal Processing** to define the physical boundaries (Wavelets, Retinex), **Self-Supervised Learning** to provide the adaptability (Non-reference losses), and **Task-Aware Constraints** to provide the purpose (Semantic preservation). This unified approach addresses the most glaring gaps in the current literature—specifically the handling of glare, noise, and temporal instability in a singular, lightweight framework suitable for the next generation of autonomous mobility.

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